CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

December 2016

EC210 Analog Communication

Date: 07.12.2016, Wednesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Describe different properties of Fourier transform with suitable examples. **[07]**

Q - 2 (a) Define amplitude modulation and derive mathematical equation for amplitude modulated wave. **[07]**

(b) Explain with the help of neat diagram and waveforms working of collector modulated class C amplifier. **[07]**

OR

Q - 2 (a) The positive R_f peaks of an AM voltage wave rise to a maximum value of 15V and drop to a minimum value of 5V. Determine the modulation index and the unmodulated carrier amplitude, assuming sinusoidal modulation. **[07]**

(b) How broadcast Amplitude Modulated (AM) transmitter transmit information signal in real scenario?-Describe its step process. **[07]**

Q - 3 (a) How receiver tracking system works? Describe various types of receiver tracking system. **[07]**

(b) Write a note on tuning range in receiver with suitable examples. **[07]**

OR

Q - 3 (a) Draw and explain the block diagram of superheterodyne receiver with suitable signal spectra. **[07]**

(b) What is sensitivity and gain in receiver? Describe the same with necessary diagram. **[07]**

SECTION – II

Q - 4 (a)

i) What is the power saving in DSB-SC-AM and SSB-SC AM? **[02]**

ii) A transmitter supplies 8 Kw to the antenna when modulated. Determine the total power radiated when modulated to 30%. **[03]**

iii) Give the methods of generating SSB-SC-AM. And mention some applications of SSBSC. **[02]**

Q - 5 (a) Explain the following terms. **[07]**

1.Thermal Noise

2.Shot Noise

(b) Define Noise Factor. Derive Friss's formula **[07]**

OR

Q - 5 (a) Explain the following terms. **[07]**

1.Johnson Noise

2.Low frequency Noise

(b) What is signal to noise ratio of a tandem connection? Draw and explain signal to noise ratio of a tandem connection with suitable diagram. **[07]**

Q - 6 (a) Explain typical preemphasis circuit with its application. **[07]**

(b) i) What is percent modulation? Explain carson's rule for FM. **[05]**

ii) Enlist the advantages of frequency modulation. **[02]**

OR

Q - 6 (a) Explain frequency modulation with waveform. Also derive mathematical equation of frequency modulated (FM) wave. **[07]**

(b) In an FM system, when the audio frequency is 500Hz and modulating voltage 2.5V, a deviation produced is 5KHz.If the modulating voltage is now increased to 7.5V, Calculate the new value of frequency deviation produced. If AF voltage is raised to 10V while the modulating frequency dropped to 250Hz, what is the frequency deviation? Calculate the modulation index in each case. **[07]**

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